B.Tech Degree VII Semester Special Supplementary Examination June 2012

EB/EC/EI 705 (D) MECHATRONICS

(*2006 Scheme*)

Time : 3 Hours Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) What are the different terms used to define the performance of transducers?

(b) What is Mechatronics? What are the key elements of mechatronics system?

(c) Explain the principles of operation of the variable reluctance stepper motor.

(d) Explain hydraulic and pneumatic power supplies.

(e) Propose a model for the metal wheel of a railway carriage running on a metal track.

(f) List the various applications of MEMS.

(g) Draw and explain the circuit of a proportional controller using op-amp.

(h) List the parameters to be considered while selecting a robot.

PART B

(4 × 15 = 60)

II. Explain the transducers used for sensing displacement, position, proximity and velocity with two examples for each.

OR

III. (a) Draw the block diagram of a generalized data acquisition system and explain.

(b) Draw a neat block diagram of data logger and explain it in detail.

IV. (a) Explain linear and rotary actuators, with the help of neat diagrams.

(b) Discuss the principle of a pilot-operated valve.

OR

V. (a) Explain the different types of ball and roller bearings.

(b) Describe the principle of the brushless dc permanent magnet motor.

VI. (a) What are the various manufacturing processes of MEMS?

(b) With a neat figure, explain the principle of fiber optic temperature sensing.

OR

VII. (a) Derive the differential equations for a permanent magnet DC motor.

(b) Explain the basic building blocks for a thermal system.

VIII. (a) With a neat block diagram explain the architecture of a PLC.

(b) Draw the block diagram of a digital closed loop control system and explain.

OR

IX. (a) Explain the different types of robot configurations.

(b) What are the different types of robot control?

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